

2. Software Project Plan

2.1 Time Schedule for Various Phases

The development of the E-Commerce Multi-Vendor Platform was planned across 10 weeks, divided into clearly defined phases. Each phase has specific deliverables that contribute toward the final working system and project report.

Weekly Project Schedule

Week	Phase	Activities	Deliverables
Week 1	Requirement Gathering	Identify user roles (Admin, Vendor, Customer); list functional and non-functional requirements; study similar platforms	Requirements document, list of features per role
Week 2	Requirement Gathering	Finalize API endpoint list; identify technology stack (Django + React + JWT); set up GitHub repository	Finalized SRS draft, tech stack decision
Week 3	System Analysis	Draw Use Case Diagram for all 3 actors; draw Level-0 and Level-1 DFDs; identify system entities and data flows	Use Case Diagram, DFD diagram drafts
Week 4	System Design	Design ER diagram and database schema; plan React component structure and folder layout; define REST API contracts	ER Diagram, DB schema, API endpoint table
Week 5	Backend Development - Core	Set up Django project; create custom User model with roles; implement JWT	Working auth API with JWT tokens

		authentication (register, login, refresh, me)	
Week 6	Backend Development - Features	Build Vendor, Product, Category, Order, and OrderItem models; implement role-based permissions; implement dashboard stats API	All Django apps with models, serializers, views, URLs
Week 7	Frontend Development - Core	Set up React + Vite project; build AuthContext, CartContext; create Navbar, ProductCard components; build Home, ProductList, ProductDetail, Login, Register pages	Public-facing React pages connected to backend
Week 8	Frontend Development - Dashboards	Build Cart, Checkout, Orders pages; build Vendor Dashboard, Vendor Products (CRUD), Vendor Orders pages; build Admin Dashboard, Admin Vendors, Admin Orders pages	All role-specific dashboards and protected routes working
Week 9	Testing & Debugging	Unit testing of Django models; API testing with Postman; integration testing of React ↔ Django; fix CORS issues, JWT refresh bugs, and vendor isolation bugs	Test case table, bug fix log
Week 10	Implementation & Report	Final deployment setup; write project report (all 9 chapters); prepare presentation slides; record 10-minute video	Completed project report + recorded video

2.2 Phase Summary

Phase	Duration	Description
Requirement Gathering	Week 1–2	Understanding the problem, defining roles, features, and technology stack.
System Analysis & Design	Week 3–4	Use Case and DFD diagrams, ER diagram, API contract design, component planning.
Backend Development	Week 5–6	Django REST Framework setup — models, serializers, views, JWT authentication, role-based APIs.
Frontend Development	Week 7–8	React SPA — all pages, contexts, Axios integration, protected routes, role-based navigation.
Testing & Debugging	Week 9	Unit tests, Postman API tests, integration tests, bug fixes.
Implementation & Report	Week 10	Final testing, report writing, presentation preparation, video recording.

2.3 Tools & Technologies Used

Phase	Tools / Technologies
Planning & Design	Draw.io (diagrams), Google Docs, GitHub

Backend Development	Python 3, Django, Django REST Framework, SimpleJWT, SQLite, VS Code
Frontend Development	React 18, Vite, Axios, React Router v6, Vanilla CSS, VS Code
Testing	Postman (API testing), Django TestCase (unit testing), Browser DevTools
Documentation & Presentation	MS Word / HTML / CSS, PowerPoint, Zoom (video recording)